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Mixed Convection Analysis in Lid-driven Cavity with Sinusoidally Curved Bottom Wall Using CNT-Water Nanofluid

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Abstract. Mixed convection in a lid-driven enclosure with a curved bottom wall has been investigated using CNT (Carbon Nanotube)-water nanofluid in this paper. The curvature of the bottom wall follows the sine function. Studies have been made with different amplitudes (λ = 0.05, 0.1, 0.15) of the sine function hence wall curvature. The curved wall at the bottom is heated and the top wall is kept at a relatively low temperature. Left vertical and right vertical surface are assumed to be adiabatic. Top wall has been moving at a constant lid velocity U_0 at right direction. Galerkin method of FEA (Finite Element Analysis) has been used to solve the governing equations. Different parameters like Richardson number (Ri = 0.1 ~ 10) at a fixed Reynolds number (Re = 100), solid volume fraction of CNT particle (φ = 0 ~ 0.09) are used to observe better heat transfer rate. Streamlines, isothermal lines and average Nusselt number plots are included to discuss the result of the investigation. A 2D plot between average Nusselt number and solid volume fraction of CNT-water nanofluid is also given to analyse heat transfer rate. It is observed that higher value of Richardson number shows better heat transfer rate. Finally, the paper concludes that better heat transfer is achieved at higher amplitude (λ = 0.15) of curved surface at higher solid volume fraction (φ = 0.09).

INTRODUCTION

The importance of mixed convection flow in a lid-driven cavity has been enhanced substantially due to their ever increasing engineering applications such as in lubrication technologies, electronic cooling, food processing, nuclear reactors [1-3], home ventilation, electronic cooling devices and solar collectors [4]. To overcome low thermal conductivity of traditional working fluids various measures have been taken over the recent few years. Using combination of fluid and metallic particles with high thermal conductivity is one of them which is called nanofluid [5]. Many subsequent studies have shown that nanofluids yield significant improvement in the heat transfer performance in a diverse range of engineering fields [6-8]. Choi et al. [9] suggested that introducing nanoparticles into the base fluid resulted in higher thermal performance for the resultant nanofluid. From the experimental work of Eastman et al. [10] it was seen that an increase in thermal conductivity of approximately 60% was obtained for a nanofluid consisting of water and 5 vol. % CuO nanoparticles. Das et al. [11] also found out that 2–4-fold increase in thermal conductivity was obtained for water-based nanofluids containing Al_2O_3 or CuO. The convective heat transfer characteristics of nanofluids depends mainly on the thermal properties of both the base fluid and the ultra-fine (<100nm) particles as well as physical properties of the base fluid, the flow pattern, flow structure and the volume fraction, dimensions and the shape of the suspended particles. Thermal performance of lid-driven enclosures using solid nanoparticles in base fluid has been investigated numerically very frequently in the past few years. Tiwari et al. [12] studied the mixed convection heat transfer performance of Cu-water nanofluid confined within a

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two-sided lid-driven square cavity with moving vertical walls kept at different constant temperatures. The results showed that the Richardson number and the direction of the moving walls had significant effects on both the fluid flow and the heat transfer within the cavity. A similar investigation was conducted by Muthtamilselvan et al. [13] on rectangular enclosure with different boundary conditions. Nikfar et al. [14] worked using the same nanofluid with wavy sidewalls. Both study showed that the addition of nanoparticles resulted in a significant improvement in the mean Nusselt number and thus enhanced the rate of convection heat transfer. Moreover, the results showed that the amplitude, wavelength and waveform of the wavy wall had critical effect on the flow and temperature fields thereby they directly affected the heat transfer performance.

The present study is driven by the aforementioned motivation. Heat transfer of nanofluid in a lid-driven cavity with a sinusoidally wavy wall at bottom is studied here. The amplitudes of the sine function are changed in order to find out which amplitude is more effective. Also Richardson number is varied in each case for better characteristics.

PROBLEM SPECIFICATION

The geometry considered in this paper is a cavity according to Fig. 1. Width and height of the cavity is denoted by L. The 2-D Cartesian co-ordinate system is defined and the effect of the gravity is shown on the negative y-axis. Top surface of the cavity is moving from left to right horizontally at a constant velocity U_0. Bottom wall is considered to follow sine function with three different amplitudes 0.05, 0.1 and 0.15 which is denoted by parameter $\lambda = a/L$.

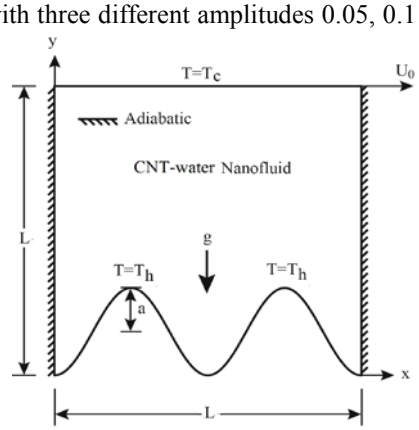

FIGURE 1. Schematic diagram for the cavity with wavy bottom wall

Top surface is kept at relatively low temperature T_c where the wavy bottom wall is uniformly heated and maintained at T_h. Left and right vertical surfaces are assumed to be adiabatic. The cavity is filled with CNT-water nanofluid. Nanoparticles and water are assumed to be in thermal equilibrium with no slip between them. The nanofluid is considered Newtonian and incompressible and the flow is assumed to be laminar. The density variation is considered according to the Boussinesq approximation. Other thermo-physical properties of the nanofluid are assumed to be constant. The thermo-physical properties of nanoparticles and water are presented in Table 1.

TABLE 1. Thermophysical properties of CNT particles and water

Property	Water	CNT
Specific heat, *Cp (J/kg K)*	4179	702
Thermal conductivity, *k (W/mK)*	0.613	3500
Thermal expansion coefficient, β *(K-1)*	21×10^{-5}	0.85×10^{-5}
Density, ρ *(kg/ m^3)*	997.1	1300

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MATHEMATICAL FORMULATION

Assumptions have been made that the nanofluid is a Newtonian fluid and the flow is steady laminar flow. Incompressible Navier–Stokes equation has been applied for the two dimensional flow. Governing equations in non-dimensional form can be written as,

$$\frac{\partial(U\gamma)}{\partial X}+\frac{\partial(V\gamma)}{\partial Y}=\frac{\partial}{\partial X}\left(\Gamma_\gamma\frac{\partial\gamma}{\partial X}\right)+\frac{\partial}{\partial Y}\left(\Gamma_\gamma\frac{\partial\gamma}{\partial Y}\right)+S_\gamma \tag{1}$$

Here, γ. Γ_γ and S_γ signify dependent variable, diffusion term and source term respectively (see Table 2)

Table 2. A summary of the governing non-dimensional equations

Equations	γ	Γ_γ	S_γ
Continuity	1	0	0
U-momentum	U	$\mu_{nf}/\rho_{nf}\alpha_{nf}$	$-\partial P/\partial X$
V-momentum	V	$\mu_{nf}/\rho_{nf}\alpha_{nf}$	$-\partial P/\partial Y+(\rho\beta)_{nf}/\rho_{nf}\beta_f\, Ra\,\Pr\Theta$
Energy	Θ	α_{nf}/α_f	0

The density of nanofluid,

$$\rho_{nf}=(1-\varphi)\rho_f+\varphi\rho_s. \tag{2}$$

ρ_{nf} is assumed to be constant. Here, φ is the solid volume fraction has significant effect on heat transfer. The thermal diffusivity of nanofluid is quite different from the conventional fluid and can be expressed by,

$$\alpha_{nf}=k_{nf}/(\rho c_p)_{nf} \tag{3}$$

$$(\rho c_p)_{nf}=(1-\varphi)(\rho c_p)_f+\varphi(\rho c_p)_s \tag{4}$$

Again the thermal expansion coefficient $(\rho\beta)_{nf}$ of the nanofluid can be found by,

$$(\rho\beta)_{nf}=(1-\varphi)(\rho\beta)_f+\varphi(\rho\beta)_s \tag{5}$$

Dynamic viscosity (μ_{nf}) of the nanofluid can be expressed by the following equation,

$$\mu_{nf}=\mu_f/(1-\varphi)^{2.5}. \tag{6}$$

Effective thermal conductivity of the nanofluid can be expressed as

$$k_{nf}/k_f=\left[k_s+2k_f-2\varphi(k_f-k_s)\right]/\left[k_s+2k_f+\varphi(k_f-k_s)\right] \tag{7}$$

Here, $k_s=$ thermal conductivity of the nanoparticles, $k_f=$ thermal conductivity of the base fluid

Following scales are used to get the non-dimensional governing equation.

$$X=\frac{x}{L},Y=\frac{y}{L},U=\frac{uL}{\alpha_f},V=\frac{vL}{\alpha_f},P=\frac{(p+\rho_f gy)L^2}{\rho_{nf}\alpha_f^2},\Theta=\frac{T-T_c}{T_h-T_c},Ra=g\beta_f L^3\frac{T_h-T_c}{\alpha_f\nu_f},Pr=\frac{\nu_f}{\alpha_f} \tag{8}$$

Here, Θ = non-dimensional temperature, Ra = Rayleigh number, Pr = Prandtl number

On the top horizontal wall: $U = 1$, $V = 0$, $\Theta = 0$. (9)

On the bottom wavy walls: $U = V = 0$, $\Theta = 1$. (10)

On the vertical side walls: $U = V = 0, \frac{\partial\Theta}{\partial X}=0$ (11)

For the bottom wavy heated surface average Nusselt number is evaluated by,

$$Nu=-\frac{k_{nf}}{k_f}\int_0^1\frac{\partial\Theta}{\partial Y}dX. \tag{12}$$

Stream function from the velocity components of a two dimensional flow can be expressed as,

$$U=\frac{\partial\Psi}{\partial Y},V=-\frac{\partial\Psi}{\partial X} \tag{13}$$

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NUMERICAL METHOD

In order to discretize the governing equations and corresponding boundary conditions mentioned in the mathematical modeling section, Galerkin method of Finite Element Analysis (FEA) has been used. The converging criterion was defined as a relative residual value of less than 10^{-6}.

Code Validation

In order to validate the numerical code presented in the present investigation, a previously studied test case is employed using the presented code. Mixed convection of Cu-water nanofluid in a lid driven cavity with sinusoidal heating on sidewalls is considered which is studied earlier by Abbasian et al. [4]. Simulation is performed using the presented code at Ri = 1, Gr = 100 for two cases of volume fraction of nanoparticles φ = 0.1 and φ = 0. From the figure it is clearly seen that the comparison of isotherm is found out to be clearly favorable.

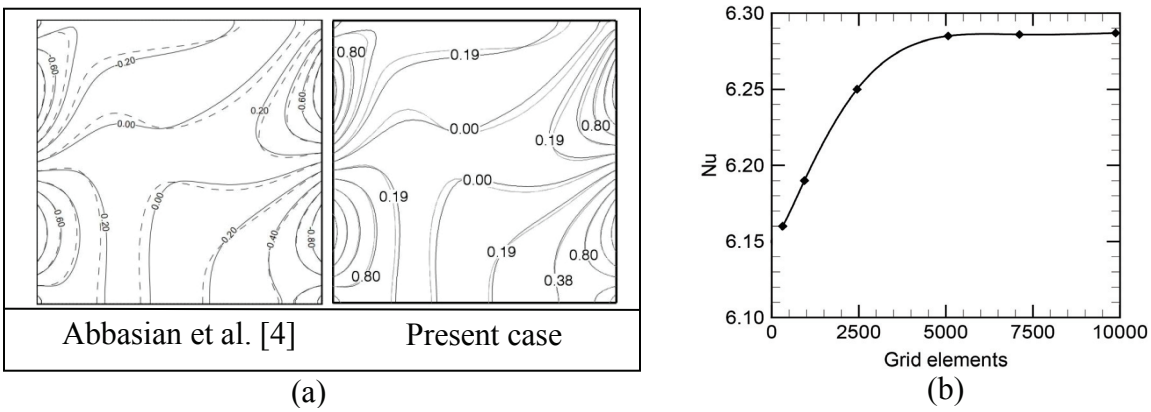

FIGURE 2. (a) Comparison of isotherm between Abbasian et al. [4] and current study at Ri = 1, Gr = 100 for φ = 0.1 (black line) and φ = 0 (gray line) and (b) Variation of Nusselt number with different grid elements at Ri = 1, Re = 100, φ = 0.1 for λ = 0.15.

Grid Independency Test

A grid independency test is performed to guarantee the optimal comprise between numerical accuracy and computational time limit. From fig. 2 (b) variation of average Nusselt number is investigated considering mixed convection for six different number (314, 940, 2452, 5067, 7114 and 9875) of grid elements at Ri = 1, Re = 100, φ = 0.1 and λ = 0.15. From number of elements 5067 no mentionable variation of Nu is observed. Further improvement of mesh size does not result in any significant change in Nusselt number. Therefore, a number of 5067 elements is selected as the ideal mesh size.

RESULT AND DISCUSSION

Effect of Amplitude on Streamline and Isotherms

From Fig. 3, by observing the streamlines for Ri = 1 and various sinusoidal wavy surface wall amplitude (λ) it can be seen that single large fully developed clockwise eddy is formed inside the cavity due to nanofluid. Based on the amplitude of the wavy wall the core position varies. As Gr = 1000, buoyancy force cannot overcome the effect of viscous force. Hence the flow inside the cavity is basically established by the moving top surface. That is the reason why the location of the core of the eddy is more near to the moving top wall than the bottom wall. From fig. 4 by observing the isotherms for a constant Ri and various λ, it can be observed that the isotherms are bundled at a greater extent near the bottom surface where the gathering rather quiet light in the interior of the cavity. Thus it indicates that a steep temperature gradients in the vertical direction in this part of the cavity, while the inner part has a very weak temperature gradients. When λ = 0.05 the isotherms cluster more into the peak region of the curve of

the bottom surface. It is due to the cavity area reduction. For the two other cases (λ = 0.1 and 0.15) the heavy cluster areas are akin at the peak areas.

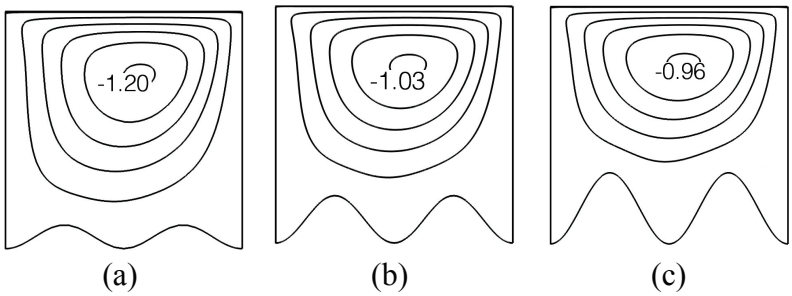

FIGURE 3. Comparison of streamline contours for (a) λ = 0.05; (b) λ = 0.1; (c) λ = 0.15 for Ri = 1, φ = 0.09 and Gr = 1000.

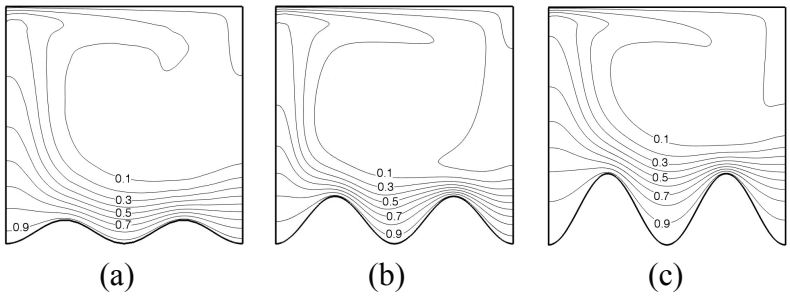

FIGURE 4. Comparison of isotherm contours for (a) λ = 0.05; (b) λ = 0.1; (c) λ = 0.15 for Ri = 1, φ = 0.09 and Gr = 1000.

Analysis of Mid-plane Vertical Velocity

Fig. 5 represents mid-plane vertical velocity, V along the x-location of the cavity. It was taken for different Richardson Number and different amplitude of wavy wall (λ). The figure indicates that the velocity increases close to the cold surface at left. It is expected due to thermal buoyancy effect as the wall creates the most temperature difference. Figure shows that when Ri is lowest, amplitude of velocity is higher at high λ. At Ri = 0.1 velocity is higher for λ = 0.15 than λ = 0.05, though the difference is not much mentionable. But as Richardson number goes up, buoyancy force becomes more dominant and amplitude of vertical velocity gradient is decreased. At Ri = 10 velocity amplitude is notably high for λ = 0.05 than λ = 0.15.

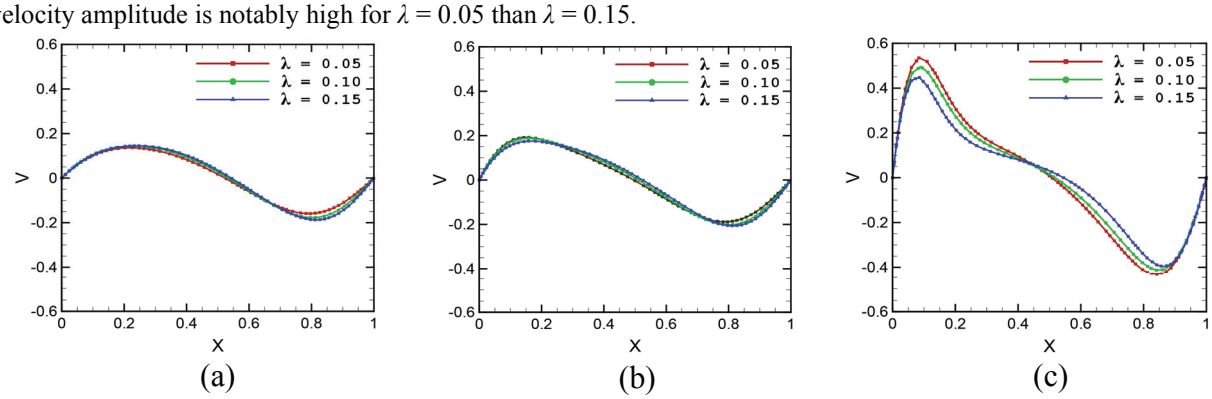

FIGURE 5. Comparison of mid-plane velocity at different λ for (a) Ri = 0.1; (b) Ri = 1; (c) Ri = 10.

Effect of Different Parameters on Average Nusselt Number

Fig. 6(a) presents the change of average Nusselt number, Nu for volume fraction φ = 0.09 with Richardson number, Ri for different λ. From the figure it can be clearly seen that Nu increases exponentially with the increment of Ri. At a same Ri, average Nusselt number is different for different λ. Nu is higher for higher λ (= 0.15) whereas

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Nu is lowest for the minimum λ (= 0.05). It is also seen that with the increment of *Ri*, increment rate of *Nu* increases as the value of λ goes down. As a result the amplitude of *Nu* difference decreases with *Ri* increase. Fig. 6(b) shows the variation of *Nu* for a constant *Ri* (=1) at φ. It is evident that, *Nu* increases linearly with the increment of volume fraction at any λ. As the volume fraction, φ is the measurement of nanoparticles quantity in nanofluid it can be said that with the increment of nanoparticle volume Nusselt number increases. Apparently at a certain φ, average Nusselt number is maximum for stronger wave amplitude (λ = 0.15) and lower as the amplitude gets lower. It is minimum at λ = 0.05. Also the difference between the amplitudes of *Nu* decreases with λ value reduction.

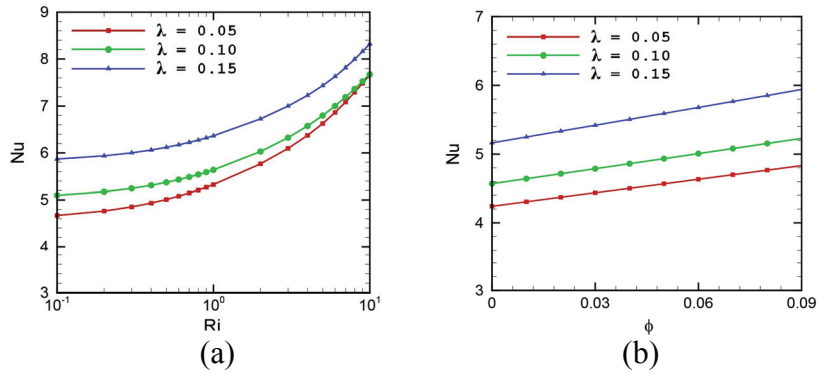

FIGURE 6. Variation of average Nusselt number for different λ with (a) Richardson number *Ri* and (b) solid volume fraction φ.

CONCLUSION

In the present study, a numerical investigation of mixed convection in a lid-driven cavity with a sinusoidally curved bottom wall has been performed using CNT-water nanofluid. Following results have been obtained from the investigation:-

- Inclusion of nanoparticles (CNT) in base fluid signifies augmentation in heat transfer performance.
- As Richardson number increases, heat transfer rate increases significantly.
- Velocity profile changes drastically for higher Richardson number and maximum amplitude of the wavy bottom wall.
- Better heat transfer is observed for maximum amplitude of wavy bottom wall.

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